



Aditya Singhal

SRN: PES1UG24AM420

+91 8979474614 | adityasinghal.aditya33@gmail.com

linkedin.com/in/aditya-singhal-57aa87312 | github.com/Its-Aditya-Singhal

B.Tech, Artificial Intelligence & Machine Learning

PES University | Class of 2028

Education

Examination	University and Institute	Year	CGPA
B.Tech, AI & ML	PES University	2024–2028	7.40 / 10 (through Sem 4)

Projects

SVX: Secure Verified Exchange | *Rust, Tauri, TypeScript, PostgreSQL* | *Personal*

getsvx.me | [GitHub](#)

- Built an open, managed format for exchanging encrypted files between organisations, deployed publicly at getsvx.me.
- Post-quantum hybrid cryptography: ML-KEM-1024 with P-384 key sealing, Ed25519 with ML-DSA-87 signatures, ChaCha20-Poly1305 streaming, and a split-key design so neither the server nor the recipient alone can decrypt.
- Rust workspace covering the format parser, crypto, client, managed server and key agent, with OIDC authorisation, a Tauri desktop app, Python and Node.js SDKs, and fuzz-tested parsers.

Tessera | *TypeScript, Chrome Extension* | *Personal*

[GitHub](#)

- Local-first extension for ChatGPT, Claude and Gemini: rules handle everyday work, and an on-device model runs only when it is worth it.
- Prompt optimiser flags underspecified prompts, asks clarifying questions and offers a rewrite with a word-by-word diff.
- Privacy layer replaces API keys, emails, phone numbers, Aadhaar, PAN and UPI IDs with placeholders; chat handoff carries a redacted summary and code blocks to another assistant.

Jarvis | *Python, TypeScript, Rust, Gemini API* | *Personal*

[GitHub](#)

- Voice-only Mac assistant in English, Hindi and Hinglish that controls the Mac, Gmail, Google Drive and Calendar through the Gemini API.
- Speaker verification with an ECAPA voiceprint from a 24-phrase enrolment: requests that read data or act need the owner's own voice.
- On-device Whisper speech recognition and Kokoro voice, with an automated pytest suite.

Codebase Archaeologist | *Python, TypeScript, RAG, tree-sitter* | *Personal*

[GitHub](#)

- Explains unfamiliar repositories (what the code does, why it exists, how it evolved) with every answer citing its source.
- Syntax-aware chunking and hybrid retrieval (semantic vectors, keyword and symbol search) over code, commits, pull requests and issues.
- Knowledge graph of files, functions, commits and authors across 10+ languages, with impact analysis and an agent mode; works with Gemini, Ollama or Claude.

Olympic Medal Prediction | *Python, scikit-learn, Streamlit* | *ML course, team of two*

[GitHub](#)

- Replicated a Stanford CS229 medal-prediction paper and extended it to Tokyo 2020 using World Bank and Kaggle data (1,650 country-Games rows, 1988–2020).
- A rolling-validation Random Forest scored RMSE 3.07 on Tokyo 2020, against 3.68 for “same as last Games” and 3.43 for linear regression.
- Showed the paper's reported 2016 error does not reproduce on held-out data; built an interactive Streamlit demo.

Satellite Communication Management System | *MySQL, MongoDB, Node.js, React* | *DBMS course, team of two*

[GitHub](#)

- Web app for scheduling ground-station passes, leasing transponder bandwidth without conflicts, billing clients and monitoring satellite health.
- BCNF MySQL schema with 14 tables, 13 triggers, 6 stored procedures, 4 functions and role-based grants; row locks prevent double booking.
- MongoDB telemetry with anomaly detection and alert rules; Express REST API and a React dashboard.

ImpactEcho | *Python, Flask, Solidity* | *HackOween (Top 4), team of two*

[GitHub](#)

- Blockchain-backed donation platform: donors get auto-created wallets and verifiable INR transactions on a custom Python ledger.
- NGO onboarding with admin approval, cause requests and notifications, plus a Foundry-based Solidity crowdfunding contract.

Technical Skills

- Languages:** C, Python, Java, JavaScript, TypeScript, Rust, SQL, HTML, CSS
- Frameworks and Libraries:** React, Next.js, Node.js, Express, Flask, Flutter, Tailwind CSS, Tauri, Streamlit, scikit-learn, pandas, NumPy
- Databases and Tools:** MySQL, MongoDB, PostgreSQL, Git, GitHub, Docker, Linux, Render; Arduino, ESP8266 (NodeMCU)

Scholastic Achievements and Certifications

- Ranked 10 in PESSAT; placed in the top 10 in four hackathons, including Code of Honor (Top 10) and HackOween (Top 4)
- Certifications: Ethical Hacking (GUVI, 2022), Dark Web (GUVI, 2022), Quantum Computing (PES University, 2025)

Key Courses

Data Structures and Algorithms, Design and Analysis of Algorithms, Operating Systems, Database Management Systems, Machine Learning, Software Engineering, System Design, Microprocessor and Computer Architecture, Digital Design and Computer Organization, Web Technologies, Agent Ops, Prompt Engineering

Positions of Responsibility and Extra-Curriculars

- Clubs:** Samarpana (Media as video editor, Operations, Event Management), Ninada Music (Logistics), Alcoding (Technical)
- TAMS event:** Core team member, Event Management (EVM)
- Soft skills:** Problem-solving, creativity, teamwork, time management, active listening